

Haruki Munechika, Taira Kobayashi

Diagram illustrating a device structure with dimensions 50 μ (width) and 2 μ (height). The structure consists of a central vertical bar labeled '9' and two horizontal bars labeled '1' and '19'.

Diagram illustrating the equivalent circuit of a neuron membrane, showing the relationship between membrane potential V_i and various conductances and voltages.

The circuit includes the following components and connections:

- Membrane Potential:** V_i (intracellular) is the potential across the membrane, indicated by a vertical arrow on the left.
- Capacitor:** C_m represents the membrane capacitance.
- Conductances and Voltages:**
 - $g_{i-1,i}$ and $g_{i,i+1}$ are conductances connecting the intracellular space to the extracellular space.
 - V_{i-1} and V_{i+1} are voltages associated with the conductances $g_{i-1,i}$ and $g_{i,i+1}$, respectively.
 - I_{ext} is the external current entering the intracellular space.
 - V_i (intracellular) is the potential across the membrane.
- Electrochemical Potentials:**
 - E_L is the potential for the leak current $\bar{g}_L$.
 - E_{Na} is the potential for the sodium current $\bar{g}_{\text{Na}} m^2 h$.
 - E_K is the potential for the potassium current $\bar{g}_{\text{K(DR)}} n$.
- Extracellular Space:** The bottom line represents the extracellular space, which is grounded.

→ **11 states per compartment** → **209** for 19; at brain scale the gates become a **memory bottleneck**.

Development of a surrogate model capable of reproducing the membrane potential response of a multi-compartment neuron model.

④ **Sample the training data**

I_{ext}

soma

record **every** compartment

V

M

N

C

$\vdots$

$\left. \begin{matrix} M \\ N \\ C \\ \vdots \end{matrix} \right\} \text{6 gates}$

Inject a **random pulse train** at the soma; record V and **6 gates**.

② Compress gates

passes through, **not** compressed

6 gates

z_1

z_2

$z \in \mathbb{R}^n$

6 voltage-dependent gates ride a low-dimensional manifold $\rightarrow z \in \mathbb{R}^n, n = 5$. **V** stays uncompressed.

③ Identify the latent ODEs

Diagram illustrating the SINDy model structure:

- Inputs: V (blue arrow) and z (green arrow) feed into the SINDy basis.
- SINDy basis: $\alpha(V), \beta(V)$ (highlighted in a red box).
- Output: $\dot{z} = \xi \Theta(V, z)$
- Latent ODEs (highlighted in a red box):

$$\begin{aligned} \dot{z}_1 &= \xi_{11} \alpha_M(V) + \xi_{12} \beta_M(V) z_1 + \dots \\ \dot{z}_2 &= \xi_{21} \alpha_N(V) + \dots \end{aligned}$$

SINDy [3] fits **only $\dot{z}$** ; $\dot{V}$ keeps **original physics**. Library is **physics-informed** own $\alpha(V), \beta(V)$.

④ **Simulate: decoder-in-the-loop**

Diagram illustrating the **Simulate: decoder-in-the-loop** step:

- Input $z \in \mathbb{R}^n$ is processed by a **decode** block (AE, reverse of 2).
- The output of the decode block is a set of **6 gates**.
- The gates feed into an **equiv. circuit $V = f(\cdot)$ unchanged** block.
- The output of the equiv. circuit is $\dot{V}$.
- The SINDy model (ξ learned in 3) takes z as input to produce $\dot{z}$.
- The **integrate (Euler)** block takes $\dot{V}$ and $\dot{z}$ as inputs to produce the next state $z(t + \Delta t)$ and $V(t + \Delta t)$.

Data

Figure 1 displays two sets of time-series plots (0 to 1000 ms) comparing raw training trajectories (top) and latent trajectories (bottom) used to fit the SINDy model.

Top Panel: Raw training trajectories.

- Plot 1 (Top):** I_{ext} (Amp) vs Time [ms]. Shows a step-like input signal.
- Plot 2:** $V(t)$ [mV] vs Time [ms]. Shows membrane potential spikes.
- Plot 3:** $gates(soma)$ vs Time [ms]. Shows gating variables for different cell compartments.

Bottom Panel: Latent trajectories used to fit the SINDy model.

- Plot 4:** V vs Time [ms]. Shows the latent membrane potential trajectory.
- Plot 5:** $g1$ vs Time [ms]. Shows the latent $g1$ trajectory.
- Plot 6:** $g2$ vs Time [ms]. Shows the latent $g2$ trajectory.
- Plot 7:** $g3$ vs Time [ms]. Shows the latent $g3$ trajectory.
- Plot 8:** $g4$ vs Time [ms]. Shows the latent $g4$ trajectory.
- Plot 9:** $g5$ vs Time [ms]. Shows the latent $g5$ trajectory.

The legend on the right indicates the color coding for the different cell compartments: M (blue), N (green), C (red), A (purple), H (brown), and B (yellow).

④ Identified latent equations

SINDy Coefficients (SymLog Scale)

$$\hat{g}_1 = -2.11d_{\text{straw}}(V) + 56.5d_{\text{straw}}(V) - 7.6d_{\text{straw}}(V) + \dots$$

$$\hat{g}_2 = 6.92d_{\text{straw}}(V) + 238.0d_{\text{straw}}(V) - 51.8d_{\text{straw}}(V) + \dots$$

- **79.6% non-zero** — accurate but **not sparse**.

20 ms, $3 \mu\text{A}/\text{cm}^2$ step: V and the 5 AE latents.

- **When** it fires is accurate (latency err **0.3 ms**, AHP timing gap **3.1 ms**).
- **How big** is systematically **underestimated** — peak **13 mV low** (amplitude gap **12.9 mV**), rise / fall rate gap **21.0 / 10.3 mV/ms** — yet AHP depth gap is only **0.28 mV**.

Amplitude sweep). **Top**: soma, as trained. **Bottom**: dendrite, unseen.

- Bursts reproduced at **both sites** for $I \geq 5$; fires **too early** near threshold.

③ Unseen periodic drive

frequency waveform (soma)

10 Hz 20 Hz 30 Hz 40 Hz 50 Hz

hyperpolarizing current (pA)

0 -50

0 100 200 300

t [ms]

Original surrogate

Pulse train, 10–50 Hz — outside training.

- Matches for $f \geq 20$ Hz; **drops later spikes** at 10 Hz.

- Latent dynamics preserve **timing**.
- Error concentrates in the **fast, steep transition** — suggests the **AE encoding ($n=5$)**, not the SINDy library, fails to preserve it.
- Replaced **all 19 compartments**, no divergence; transfers to **unseen site · frequency**.
- **Core finding**: SINDy succeeds **not because we compressed** (n : 6→5) — ⑤ shows the **same result uncompressed** ($n = 6$), so it is the **coordinate change itself** that makes gate dynamics learnable.
- Cost reduction follows **only if** compression matters — so it remains **unproven**: ξ is still dense (**79.6%** non-zero).

→ **Future**: sparsify ξ , push n down, test on other channel models.

Code — github.com/MunehikaHaruki/SINDyNeuroSurrogate

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